

THEODOROS PSALLIDAS

Senior Software Engineer | Agentic Platforms & Machine Learning

✉ theopsall@gmail.com 📞 +30 6955648690 📍 Athens, Greece 🔗 Website in tpsallidas 📧 theopsall 🎓 Scholar

Senior engineer who ships production agentic systems: LangGraph orchestration, RAG, human-in-the-loop workflows and cross-service authorization (OpenFGA) at ProxyFoods. Since 2019 I have built full-stack and machine-learning products end to end, from IoT device fleets to multimodal video summarization. PhD candidate with 8 peer-reviewed papers. Python, TypeScript, FastAPI, React, Azure.

EXPERIENCE

Senior Software Engineer

ProxyFoods

📅 February 2025 – Present 📍 Remote, Greece

- AI-powered formulation platform for CPG food companies.
- Built the core formulation engine: nutritional analysis, cost calculation, sensory profiling and mass-balance validation across a microservices architecture.
 - Shipped agentic recipe generation with LangGraph multi-agent orchestration, RAG over food-science knowledge and human-in-the-loop ingredient matching, traced with Langfuse.
 - Designed cross-service authorization with OpenFGA (Google Zanzibar model) and multi-tenant RBAC, plus the shared auth, tracing, logging and exception libraries every service uses.

Stack: React 19, TypeScript, FastAPI, Python, LangGraph, OpenAI, Langfuse, OpenFGA, SQLAlchemy, Azure SQL, Docker, Azure, OpenTelemetry

ML Engineer & Software Developer

MyTripMyWay

📅 June 2025 – March 2026 📍 Greece

- AI-driven personalised tourism platform (University of Thessaly, University of the Aegean and industry partners).
- Designed the recommendation engine: six interpretable travel personas, direction-aware itinerary optimisation and spatial-temporal feasibility modelling (Google Maps, GraphHopper).
 - Built full-stack features on a multi-service architecture: grid-based POI aggregation (Google Places, Foursquare), booking and Stripe payment flows, and an interactive map-based itinerary UI.

Stack: Next.js, React, MobX, Tailwind CSS, Leaflet, FastAPI, Python, SQLAlchemy, MariaDB, Docker, Google Maps, Stripe

Full Stack Software Engineer

Behavioral Signals

📅 March 2024 – February 2025 📍 Remote, Greece

- Emotion-aware voice AI: PAIR call-centre routing and audio deepfake detection.
- Shipped agent-matching features and improved analytics, monitoring and scalability of real-time routing.
 - Designed, built and owned the audio deepfake-detection application end to end, from models and inference to deployment.

Stack: React, FastAPI, Django, Docker, Ansible, Azure, AWS, PostgreSQL, Milvus, MLflow

EDUCATION

Ph.D. Candidate, Computer Science

University of Thessaly

📅 2021 – Present

Machine-learning methods for automated video summarization.

M.Sc. in Data Science

University of Peloponnese

📅 2019 – 2021

B.Sc. in Computer Science

University of Thessaly

📅 2013 – 2018

SKILLS

Programming

Python · TypeScript · JavaScript · Java · C

Agentic & ML

LangGraph · RAG · OpenAI · PyTorch · TensorFlow · scikit-learn · FAISS · MLOps

Backend

FastAPI · Django · Node.js · SQLAlchemy

Frontend

React · Next.js · Angular · Tailwind CSS

Data & Infrastructure

PostgreSQL · Milvus · MSSQL · Docker · Azure · AWS · OpenFGA

LANGUAGES

Greek (native) English (fluent)

Machine Learning Engineer

NCSR Demokritos

📅 December 2019 – February 2024 📍 Remote, Greece

Computational Intelligence Lab: Museek (music monitoring) and Enorasi (video analysis) projects. Held in parallel with Optechain.

- Built a music copyright monitoring system: inference infrastructure, containerised services and pipelines deployed to IoT and Raspberry Pi fleets with Balena.
- Migrated models from classical ML to deep learning and moved to FAISS vector search, shrinking the PostgreSQL footprint while improving detection accuracy and latency.
- Built video summarization and multimodal analysis models (PyTorch, scikit-learn), improved them through architecture and feature-fusion experiments, and extended the benchmark with a less subjective labelling pipeline.

Stack: PyTorch, scikit-learn, FAISS, FastAPI, React, PostgreSQL, Docker, Balena, AWS, MLflow

Lead Software Engineer

Optechain

📅 October 2020 – February 2024 📍 On-site, Argroupoli, Greece

Full-time, in parallel with the Demokritos role.

- Led frontend (React, Angular) and backend (Node.js, C#) for digital signage (Tizen, webOS) and EV-charging stations.
- Ran the Azure and Docker stack with MSSQL and managed device-fleet updates (Balena, Mender) to improve rollout safety and up-time.

PUBLICATIONS

8 papers on Google Scholar, 5 as first author

- Video summarization based on feature fusion and data augmentation, *Computers*, 2023
- Summarization of user-generated videos fusing handcrafted and deep audiovisual features, *SMAP*, 2022
- Multimodal video summarization based on fuzzy similarity features, *IEEE IVMSP*, 2022
- Multimodal summarization of user-generated videos, *Applied Sciences*, 2021
- ARCHEO: a dataset for sound event detection in areas of touristic interest, *SMAP*, 2020

CERTIFICATIONS

Back End Development and APIs

[freeCodeCamp](#)

📅 2022

CCNA Routing and Switching (course)

[University of Thessaly](#)

📅 2016 – 2017

ACHIEVEMENTS



Code IoT Hackathon

First place, “Code the IoT Hackathon of the MaTHiSiS Project”, University of West Attica